

Short Answer-Drawing-Labeling

All Answers should be on the Answer Sheet for this section. Marks on this question packet will be ignored. 2 points each unless otherwise noted. Helpful equations:

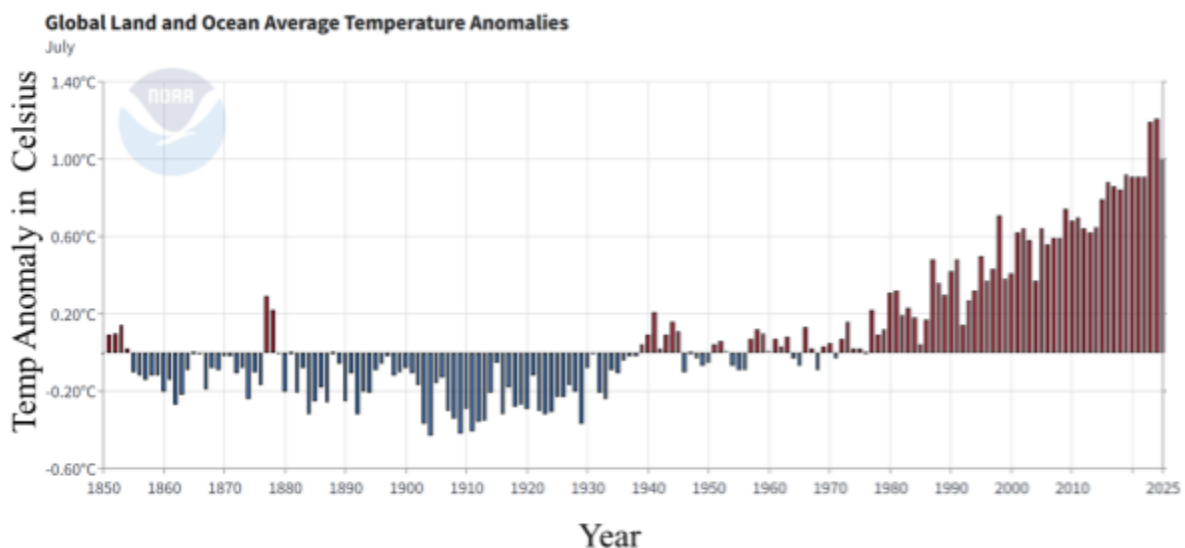
- Hardy Weinberg Equilibrium Equation $p^2 + 2pq + q^2 = 1$
- Allele frequency = $\frac{1}{2}(\text{heterozygous}) + \text{homozygous}$
- The Nernst equation is:
- $\Delta E_j = 62mV(\log \frac{C_{jo}}{C_{ji}})$

1-2. Flying lemurs have striped tails. The pattern of stripes is controlled by the expression of a protein called stripey, which makes dark fur. In most lemurs, it is turned on and off during development (hence stripes). In some lemurs, stripey is expressed all the time because of a mutation in the promoter region. These animals have all black tails. In another group of lemurs, the protein has a point mutation and doesn't function altogether, so the lemurs have white tails. If a population only had these versions, how many alleles and how many genes for stripey would be present?

The average global temperature for the period below is **13.9 °C**. The chart below shows the temperature **anomaly** for the last 175 years.

3. What was the **average global temperature in 2025**?

4. Evidence suggests that the release of what gas by humans has led to the increase in anomaly seen below?



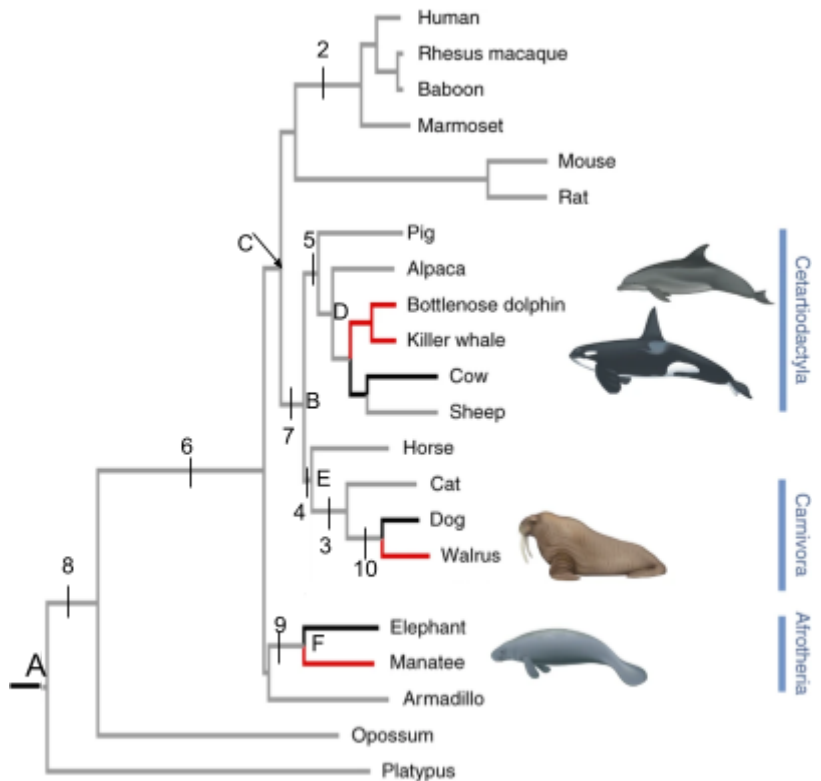
5-6. Draw two action potentials. Number 5 should be the control. **Label a-threshold; b-when Na voltage-gated channels are open; c-when K voltage-gated channels are open.** (4 points) Draw a second action potential (6) in the presence of TTX, a potent inhibitor of the **opening** of sodium voltage-gated channels (3 points). **Label the axes.**

7. The curve on the answer sheet shows the saturation of hemoglobin in adult humans. Birds that fly at high altitudes need to be able to saturate their blood even when the atmospheric pO_2 is low. They still need to release it to their cells, though. What might a hemoglobin saturation curve look like for these birds? **Add that and label it “A.”** Hemoglobin has a very high affinity for carbon monoxide. When CO is present, hemoglobin remains saturated with oxygen down to a very low $pO_2 < 5$ mmHg. **Draw a hemoglobin saturation curve in the presence of CO. Label it “B.”**

8. Pollen is the male gametophyte in plants. How many copies of each chromosome does each of its nuclei have?

Use the phylogeny at right to answer the next five questions on the answer key. **Numbers represent traits, letters represent nodes.** **Answers are either numbers, letters, or the names of animals**

9. Most recent common ancestor of the sheep and the walrus. **Letter**
 10. Synapomorphy for the carnivora **Number**
 11. Outgroup **Name**
 12. Ancestral trait for carnivora that's not present in the cetartiodactyla **Number**



Two populations of squirrels live on the UMass campus, one near the Franklin Dining Commons and one near the Worcester Dining Commons. There are two alleles for fur color in these squirrels, B and b. Homozygous BB animals are black, heterozygous Bb animals are brown, and homozygous bb animals are grey. The genotype ratios are at the right. Answer the questions on the answer sheet

13. Is one of the alleles dominant? How do you know? (<5 words) 3 pts

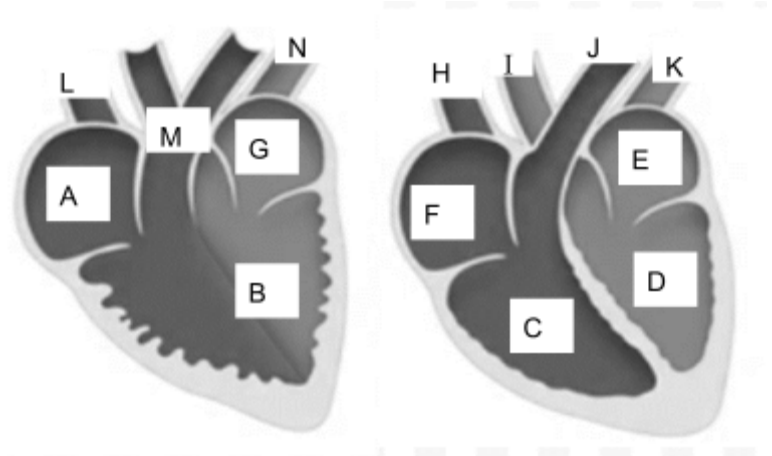
14. What is the observed frequency for the b allele in Franklin?

15. What is the expected genotype frequencies for gray squirrels in Worcester? (0.5 each)

	Black	Brown	Grey
Franklin	0.36	0.48	0.16
Worcester	0.4	0.3	0.3

16. Are the alleles in Hardy-Weinberg equilibrium in each population? **Yes or No only.**

On the right are two different hearts with letters denoting various features. Match the descriptions below with the correct feature, then fill that letter in on the answer sheet. (1 point each)



17. **Chamber** with the highest pO_2 in an amphibian heart

18. **Ventricle** with oxygenated blood in the mammalian heart

19. **Vessel** with the highest pressure in the amphibian heart

20. **Chamber** with the **lowest oxygen** in the amphibian heart

21. **Vessel** with blood returning from the body in the mammalian heart

22. What molecule causes myosin to “let go” of actin. One word? (2 pt) (muscle contraction figure at the end of the exam)

23. We looked at simulations of allele frequency change in class. A single allele from a pair of alleles was represented over several hundred generations in **five separate populations**. Draw a graph showing what you would expect to happen in **five populations over 500 generations** in a **small population** where there's an advantage to the **heterozygous genotype**. Remember the allele should start at roughly 0.5 frequency. **Remember to label your axes.** (3 point)

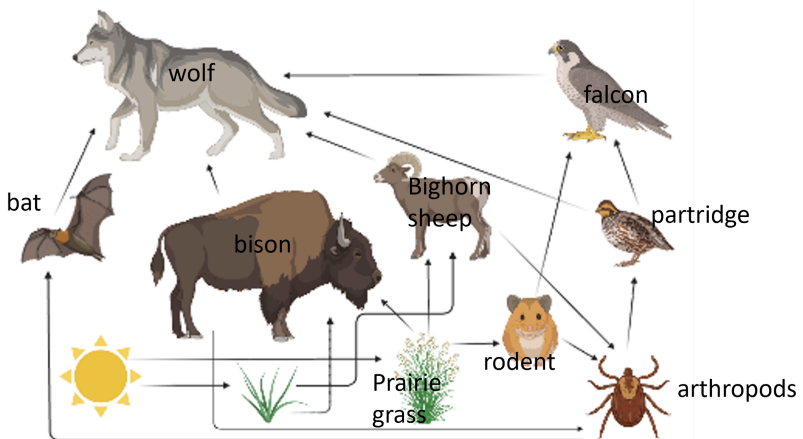
24. What type of pressure makes the xylem operate?

25. How does the xylem generate this pressure? (3 pts)

Considering the trophic web shown at right:

26. Which organism could be a dominant species?

27. What might happen to the rodents if the keystone species were removed? (< 5 words)



A mammalian cell has the following internal and external concentrations of sodium and potassium:

Ion	Intracellular mM	Extracellular mM
K ⁺	150	5
Na ⁺	5	100

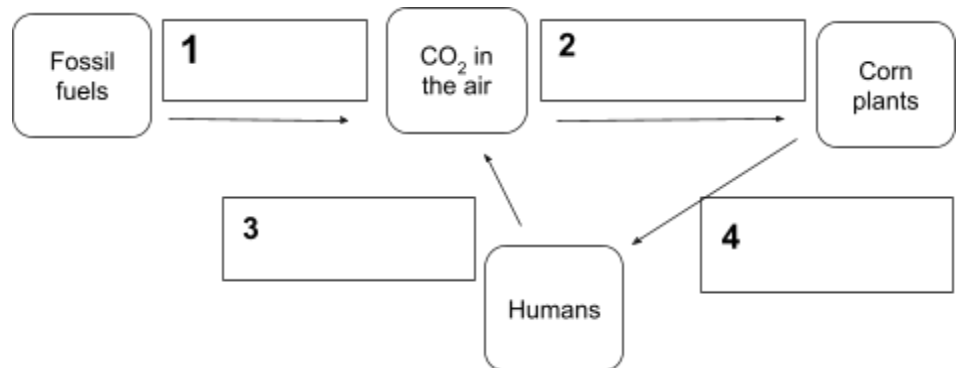
28. What is the Nernst Potential for each ion?

29. If this cell has typical ion permeabilities, approximate the membrane potential

30. What is one thing relevant to evolution/physiology or ecology that we didn't study, but that you wish we had? (1 point)

Fill in your bubble sheet on the scantron and bubble in the letters and numbers. Each question is worth 2 points unless otherwise specified

1-4. Considering the carbon cycle on the answer sheet, what processes connect the boxes shown? (Each arrow shows the movement of carbon from one sink to another. For instance, CO₂ from the air “dissolves” into water – you would write “dissolves” to connect the air and the water. (1 pt each)



A=photosynthesis
 B=Combustion
 C=Consumption
 D=Cellular respiration

5-8 Identify each as **A**: cellular pathogen, **B**: an abiotic pathogen, **C**: adaptive immune system, **D**: innate immune system, **E**: or none of these (1 point each)

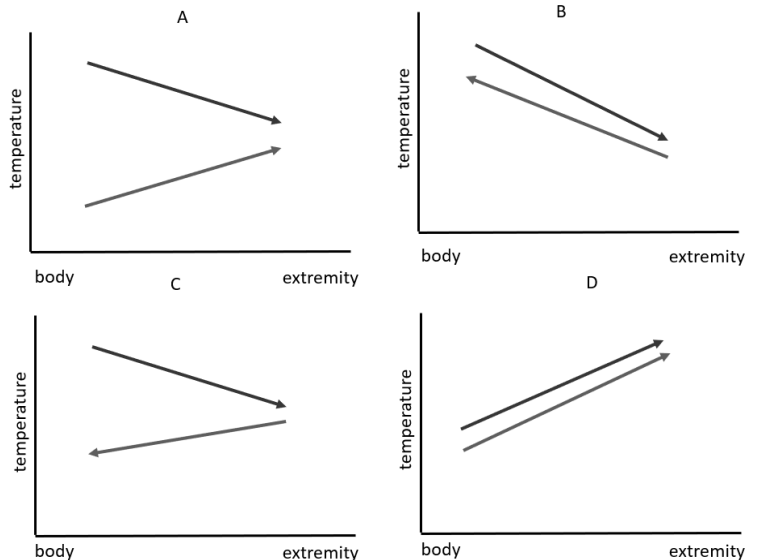
5. Influenza virus
6. Mucus membranes in lungs
7. Antibody
8. Infectious Bacteria

Snowshoe hares (rabbits) turn white in the winter. Shortening days cause their fur to change. In captivity, their fur color can be changed by altering the day length without changing the temperature.

9. The ability of this **species** to change fur color in the winter is an example of _____ because it happens when the weather changes and can be reversed in the spring.

- A. Genetic Drift
- B. Gene Flow
- C. Directional selection
- D. Adaptation
- E. Acclimitization

10. Which graph accurately represents the blood flow to an elephant's ear if they are trying to dissipate heat to the environment?

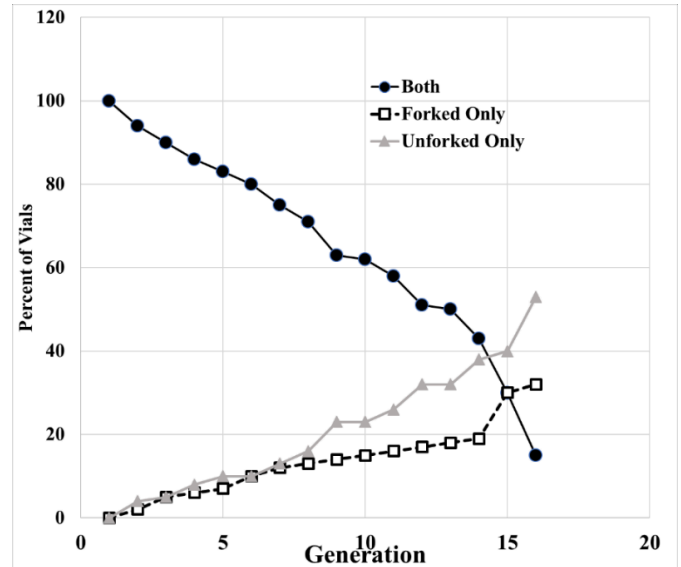


11. What does the biological species concept use as the primary criterion for determining species boundaries?

- A. geographic isolation
- B. niche differences
- C. gene flow
- D. morphological similarity
- E. molecular (DNA, RNA, protein) similarity

In the data at right, 100 groups of 8 flies each with 0.5 frequency for both forked and unforked hair alleles were placed in separate vials. Neither allele has any selective advantage. Each generation (flies have lots of offspring), **4 males and 4 females** from each vial were **randomly** chosen to start the next generation. At the end of the experiment, what can you say about the allele frequencies in the three subsets of vials?

12. If the researchers had made sure that half the flies they chose to start **each generation** had the forked allele and half lacked the forked allele (unforked), none of the vials would be “forked only.” (A=true B=false)



Describe the following as (A) analogous or (B) homologous traits using the phylogeny at right. “e” shows where eyes evolved, and P shows where the P genes evolved.

13. Eyes in Annelids and Molluscs.

14. Many beetle species have evolved the ability to subsist almost entirely on dung. Though their common ancestor didn’t share this trait. Dung beetles are therefore:

- Monophyletic
- Polyphyletic
- Paraphyletic

15. In which of the bulk flow systems below is the pressure of the blood going into the systemic capillaries significantly lower than the pressure of the blood going into the gill or lung capillaries?

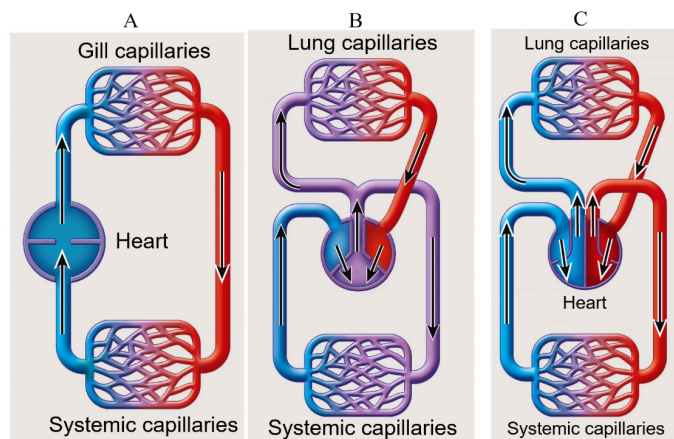
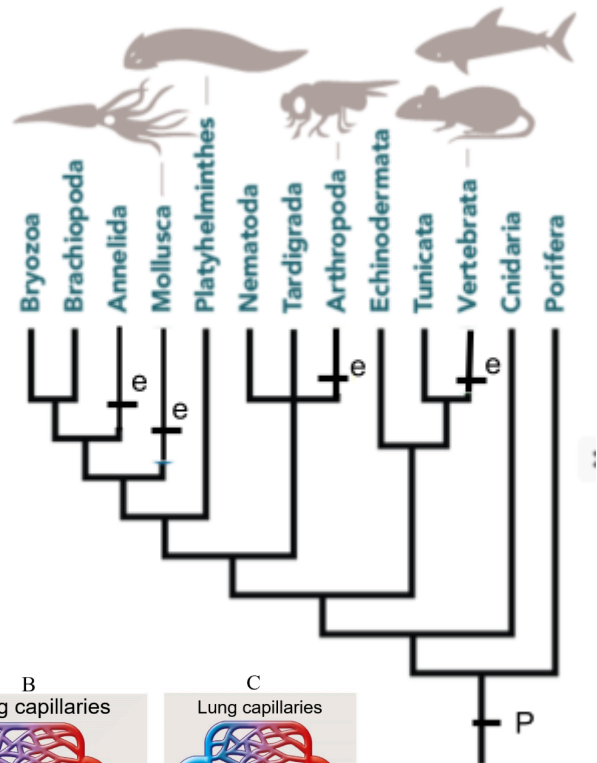


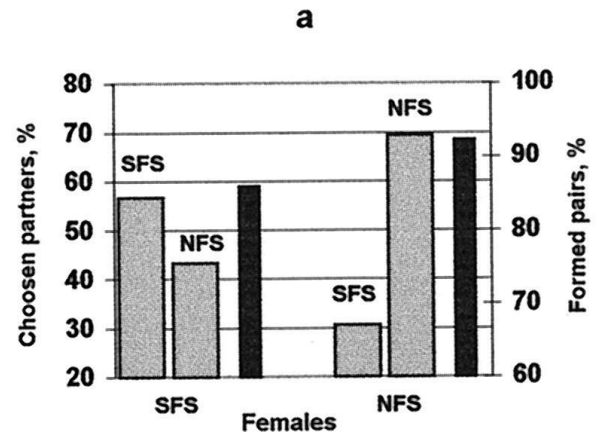
Figure 39.19b
Biology: How Life Works
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Figure 39.20b
Biology: How Life Works
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Figure 39.21b
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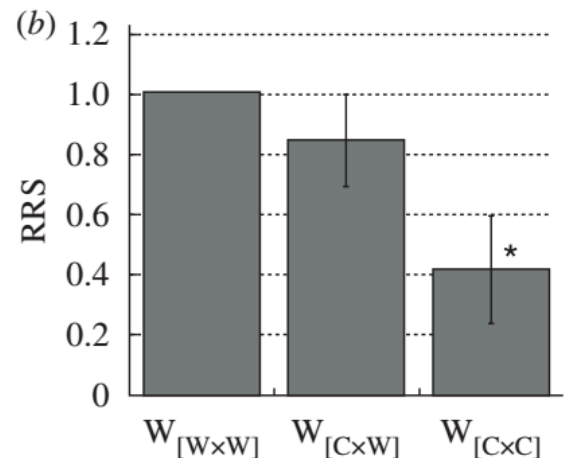


Two morphologically identical *Drosophila* populations live on opposite slopes of a valley. The environment is very different even though they are very close. **In a laboratory experiment**, flies from the north (NFS) were placed in vials with flies from the south (SFS). In the data at right you can see the female choices. On the left of the graph are females from the south and which males they chose. On the right of the graph are northern females. Black lines represent the number of successful pairs.



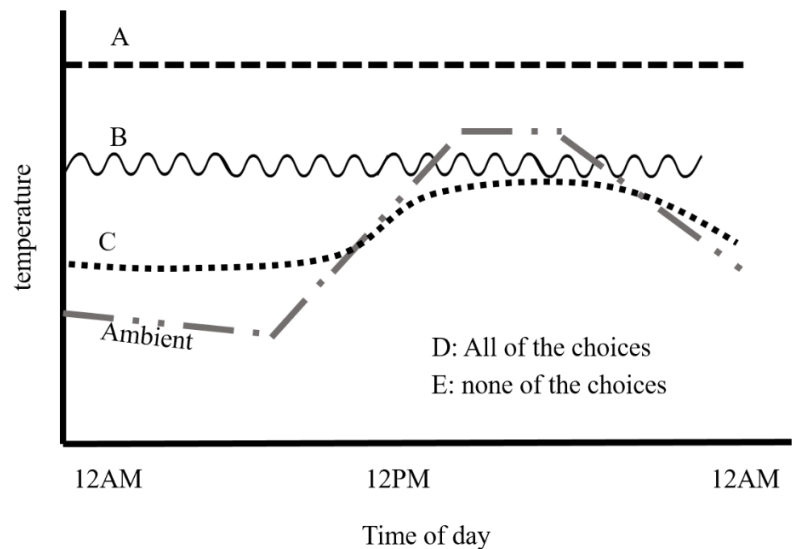
16. This is a laboratory experiment showing
- Adaptive radiation
 - Genetic Drift
 - Mechanical reproductive isolation
 - Sympatric speciation
 - Allopatric speciation

In class we looked at the data at right. It represents the relative reproductive success (RRS) of trout in the wild who had either two wild parents (WXW), one wild and one captive bred parent (WXC), or two captive bred parents (CXC). Evaluate each of the following statements.



17. This is an example of artificial selection for traits that give a **fitness advantage in the wild**. A=true, B=false
18. Gene flow between the two populations is leading to lower fitness for the whole population in the next generation. A=true, B=false

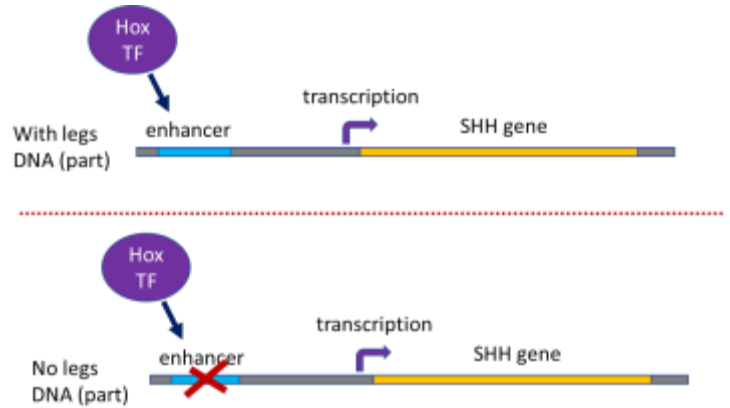
In the chart at right three different animals core body temperatures are represented along with the ambient temp. Ambient is a grey line with dashes and dots. Use the chart to answer the next two questions



19. Which is an ectotherm?
20. Which uses homeostatic control of its temperature?

21. In pythons, a 17 base pair region is missing from an enhancer region for the *Shh* gene. This enhancer is a target for HoxD binding. In mice when this same enhancer region is removed it leads to limb loss. What happens to HoxD when the 17 base pair region is removed?

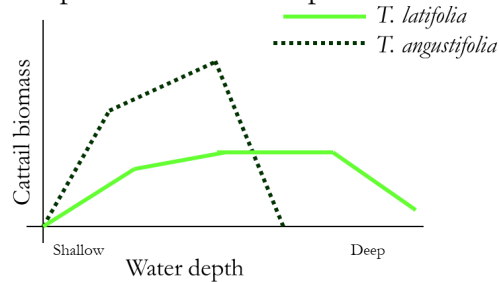
- A. It is expressed at a lower level
- B. It cannot bind the SHH protein
- C. It cannot bind the SHH enhancer
- D. It is expressed at a higher level



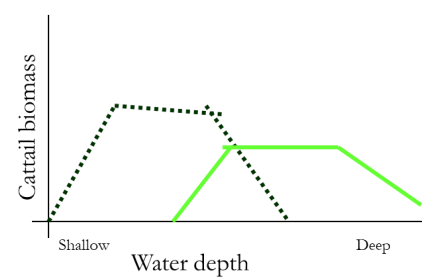
Researchers were studying two species of cattail (a plant species that grows along the water's edge) and comparing cattail growth along a gradient of water depth.

Typha angustifolia dominates in shallow water, and *Typha latifolia* dominates in deep water (see below at right). Researchers wanted to determine if competition or water depth was limiting the growth of either species. They conducted transplant experiments: they planted each species on its own in both deep and shallow water and compared them to the natural populations. The results of the planting experiments are shown above—note that even though it shows the biomass on one graph, the plants were in separate plots.

Transplants without competition

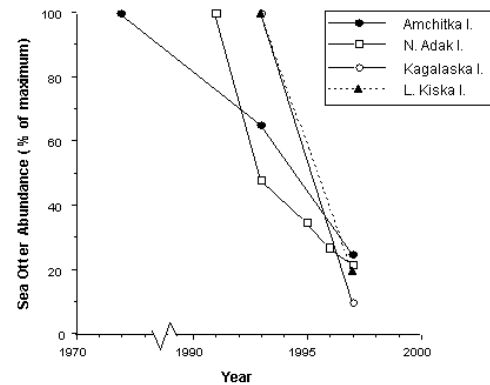
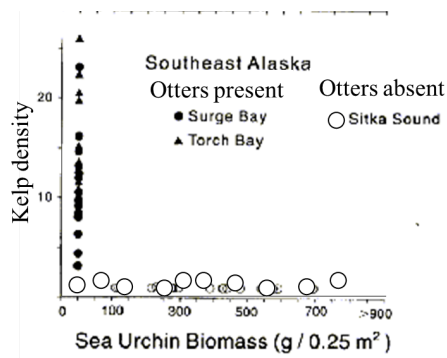


Natural populations



22. The realized niche of which species is closest to its fundamental niche?

- A. *T. latifolia*
- B. *T. angustifolia*



The data above and at left are from surveys taken on kelp (**a photosynthesizing algae**), sea urchin (**an animal which eats kelp**) and otters (**which eat sea urchins**). Sea otter populations are shown at right. Use these data to answer the questions below

23. Sea urchins are a dominant species in habitats with no otters. (A=true, B=false)
24. Sea urchins act as keystone species controlling kelp density. (A=true, B=false)

Muscle Contraction

